

From Prediction Uncertainty to Conformalized Distance Fields for Safe Motion Planning

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Abstract—Safe motion planning in dynamic environments requires reasoning about the uncertainty of predicted obstacle motion without sacrificing real-time performance. We introduce a functional conformal prediction (FCP) framework that produces a high-confidence lower bound of the predicted signed distance field (SDF) at the level of the spatial *field* rather than at individual points or trajectories. The residual SDF is represented in a data-driven functional basis via functional principal component analysis, and a Gaussian-mixture inductive conformal procedure yields a distribution-free upper envelope of the score. A key feature is an offline–online decomposition in coefficient space: an accurate envelope is fitted offline, while online it is refined by a lightweight adaptive functional conformal (AFCP) update acting only on a low-dimensional coefficient vector. We embed the conformalized lower-bound SDF as a tightened safety constraint in a sampling-based model predictive controller and establish asymptotic closed-loop safety, both under exchangeability and, via the online adaptation, under distribution shift. Experiments on the ETH-UCY pedestrian benchmarks and a dense 3D quadrotor navigation task with hundreds of dynamic obstacles show that FCP-MPC attains a favorable balance of safety, feasibility, and efficiency, while keeping per-step computation far below online uncertainty-reasoning baselines.

Index Terms—Conformal prediction, motion planning, model predictive control, collision avoidance, signed distance fields, uncertainty quantification, safe navigation.

I. INTRODUCTION

efficient motion planning in uncertain environments with safety & efficiency guarantees

conformal prediction for safe motion planning [1]–[7]

- representation perspective (environmental geometries often admit low-dimensional representations)
- recent worst-case definition of score functions for motion planning are intractable, thus requiring severe approximation in practice, e.g. [4], [7].

A. Notation

For $f, g \in L^2(\mathcal{X}, dx)$, we denote their inner product by $\langle f, g \rangle = \int_{\mathcal{X}} f(x)g(x)dx$.

II. RELATED WORKS

A. Navigation in Dynamic Environments: Perception and Prediction

Safe operation in dynamic environments has classically been decomposed into estimating the evolving occupancy of the world and planning against it. A large body of work tracks

moving obstacles by coupling simultaneous localization and mapping with multi-object tracking [8], [9], or by maintaining dynamic occupancy representations through particle-based grids [10], [11] and random-finite-set filters [12], [13]. Among these, continuous particle-based occupancy mapping has been pushed to real-time dense settings [13], reflecting a broader trend toward distance- and occupancy-aware spatial representations. In parallel, signed distance fields—learned [14], [15] or fused online from depth sensing [16]—have become a favored geometric layer because of their smooth, queryable structure. Our formulation inherits this representational viewpoint: rather than reasoning over raw occupancy, we operate directly on the predicted signed distance field, and we further treat the prediction error of that field as the object to be quantified. On the forecasting side, learned trajectory predictors such as Trajectron++ [17] supply the multi-step obstacle motion that our planner consumes, and their residual errors are precisely what a safety layer must account for.

B. Risk-Aware Motion Planning

A complementary line of research embeds uncertainty directly into the planning objective or constraints. Chance-constrained formulations bound the probability of collision and have been studied extensively, from convex chance-constrained path planning with obstacles [] to probabilistic collision checking under explicit chance constraints [] and motion planning in dynamic, uncertain environments []. Because exact verification of collision constraints against non-convex moving obstacles is generally intractable, these methods typically approximate the free space or the obstacle set, e.g., by unions of convex regions [], or accept conservative encodings of the nonconvex feasible set []. Risk-sensitive planners instead optimize coherent risk measures such as the conditional value-at-risk [], and distributionally robust approaches hedge against the worst-case distribution within an ambiguity set; the distributionally robust risk map [], for instance, certifies CVaR-type safety even when the obstacle motion distribution is only imperfectly learned. These methods deliver strong guarantees but generally require an explicit (often parametric or moment-based) model of the prediction uncertainty, and the worst-case machinery is frequently solved online, which can be costly in dense scenes. Our work targets the same safety semantics while pushing the expensive statistical computation offline and keeping only a lightweight coefficient-space update online.

- robust control: DRO
- navigation in dynamic environments
 - perception (localization, mapping, detection, tracking, ...) & prediction in dynamic environments

- * SLAM + MOT [8], [9]
- * particle-based [10], [11]
- * random finite sets [12], [13]; based on [18]
- * SDF learning [14], [15]; mostly in 3d; may be irrelevant to ours
- risk-aware motion planning [19]–[22]: no safety guarantee, exact formulation is often intractable; require substantial approximations

III. PRELIMINARIES

Our machinery builds on conformal prediction (CP), a distribution-free framework that turns a heuristic uncertainty score into sets with finite-sample coverage. We recall the two ingredients we rely on; the functional extension specific to signed-distance fields is developed in Section V.

a) *Split conformal prediction.*: Given exchangeable samples $\{z_n\}_{n=1}^N$ and a nonconformity score $s(\cdot) \in \mathbb{R}$ (larger meaning a worse fit), fix a target miscoverage level $\alpha \in (0, 1)$ and set $\hat{q} = \text{Quantile}_{1-\alpha}(\{s(z_n)\}_{n=1}^N \cup \{+\infty\})$. For a fresh exchangeable sample z_{N+1} ,

$$\mathbb{P}[s(z_{N+1}) \leq \hat{q}] \geq 1 - \alpha, \quad (1)$$

with no assumption on the data distribution beyond exchangeability [23]. We use this offline to obtain a high-confidence upper bound on the signed-distance prediction error.

b) *Adaptive conformal inference.*: When the stream is non-exchangeable (e.g. under distribution shift), the marginal guarantee (1) may fail. Adaptive conformal inference (ACI) [24] instead enforces a *long-run* guarantee by updating the threshold online from the observed miscoverage: with step size $\gamma > 0$ and $\text{err}_t = \mathbb{I}[s_t > \hat{q}_t]$,

$$\hat{q}_{t+1} = \hat{q}_t + \gamma(\text{err}_t - \alpha), \quad (2)$$

which drives the empirical miscoverage $\frac{1}{T} \sum_t \text{err}_t$ to α for an arbitrary, even adversarial, sequence. Our online envelope update (29) is a coefficient-space, functional instance of (2), and underlies the robust safety guarantee of Theorem 2.

IV. PROBLEM FORMULATION

Let $\mathcal{Q}$ be the configuration space of a robot, and $\mathbf{x} = \mathbf{x}(\mathbf{q}) \in \mathcal{X} \subseteq \mathbb{R}^d$ the position in the world corresponding to $\mathbf{q} \in \mathcal{Q}$ (i.e., the forward kinematics of the robot), where $d = 2$ or $d = 3$. We also use the notation $\mathcal{A} = \mathcal{A}(\mathbf{q}) \subseteq \mathcal{X}$ to denote the footprint of the robot (which is a closed subset of $\mathcal{X}$) whose configuration is $\mathbf{q}$. Defining a control input as $\mathbf{u} \in \mathcal{U}$, the dynamics of the robot is given as the equation

$$\mathbf{q}_{t+1} = f(\mathbf{q}_t, \mathbf{u}_t), \quad t = 0, 1, \dots$$

We specifically consider a *dynamic* environment where $t = 0, \dots, T$ is the time window and $\mathcal{O}_t \subseteq \mathcal{X}$ is the union of *dynamic* obstacles at t (which is often represented as an occupancy map in practice), which is *completely unknown*. $\{\mathcal{O}_t\}_{t=0}^T$ is modeled as a set-valued random process, where the randomness of the process embraces *environmental variations*. It is also assumed that each $\mathcal{O}_t$ is independent of $\mathbf{q}_0, \mathbf{u}_0, \dots, \mathbf{q}_t$, which indicates that the motion of the robot

does not affect the environmental occupancy $\mathcal{O}_t$.¹ We also write $\mathcal{A}_t := \mathcal{A}(\mathbf{q}_t)$. Finally, to model a limited perception range, we follow the formulation of [25] and introduce a perception map $P : \mathcal{X} \times \mathcal{C}(\mathcal{X}) \rightarrow \mathcal{M}_{\text{partial}}(\mathcal{X})$, where $\mathcal{C}(\mathcal{X})$ is the collection of all closed subsets of $\mathcal{X}$ (each $\mathcal{C} \in \mathcal{C}(\mathcal{X})$ being a union of obstacles, i.e. the *unknown* occupancy of the world), and $\mathcal{M}_{\text{partial}}(\mathcal{X})$ is the set of all measurable *partial* functions on $\mathcal{X}$,

$$\mathcal{M}_{\text{partial}}(\mathcal{X}) := \{D : \mathcal{Y} \rightarrow \mathbb{R} \mid \mathcal{Y} \subseteq \mathcal{X}, D \text{ measurable}\}.$$

Intuitively, $P(x, \mathcal{C})$ is the signed distance field (SDF) sensed from configuration x when the true geometry is $\mathcal{C}$; in practice it is realized as a truncated SDF [16], which we prefer to a binary occupancy map for its smooth, distance-aware structure. The use of *partial* functions accounts for the finite sensing range. Let $\mathcal{C}_t$, $t \geq 0$, denote the time-varying occupancy, whose dynamics reflect the motion of the environment. At time t the robot at $x_t = x(\mathbf{q}_t)$ receives the observation $\hat{D}_t = P(x_t, \mathcal{C}_t)$ with domain $\mathcal{D}_t$, and the region sensed up to time t is

$$\mathcal{D}_{\leq t} := \bigcup_{\tau \leq t} \mathcal{D}_\tau.$$

On $\mathcal{D}_{\leq t}$ the observations are fused into an SDF estimate $D_{\leq t} : \mathcal{X} \rightarrow \mathbb{R}$, which is set to 0 on the unexplored region $\mathcal{X} \setminus \mathcal{D}_{\leq t}$. Given $\mathcal{O}_t$, We define the *distance function* D_t associated with $\mathcal{O}_t$ as

$$D_t(\mathbf{x}) := D(\mathbf{x}, \mathcal{O}_t) = \inf_{\mathbf{x}' \in \mathcal{O}_t} \|\mathbf{x} - \mathbf{x}'\|, \quad \mathbf{x} \in \mathcal{X}.$$

Since the future occupancy $\mathcal{O}_{t+i}$ for $i > 0$ is unobservable at time t , it must be estimated from historical and contextual data, as modern prediction models do. We denote the i -step-ahead prediction made at t by $\mathcal{O}_{t+i|t}$. For safe motion planning, we are interested in correctly estimating the (signed) distance function at $t + i$:

$$D_{t+i}(x) := d(x, \mathcal{O}_{t+i}), \quad x \in \mathcal{X}$$

and its i -step ahead prediction be:

$$D_{t+i|t}(x) := d(x, \mathcal{O}_{t+i|t}), \quad x \in \mathcal{X}$$

We are primarily interested in solving the following problem:

Problem 1 (Motion Planning with Probabilistic Safety Guarantee). *Design a policy π (inputs & outputs; inputs: partial observations / outputs: $\mathbf{u}_t$) that solves*

$$\begin{aligned} \min_{\pi} \quad & \sum_{t=0}^{T-1} \ell(\mathbf{q}_t, \mathbf{u}_t) + \ell(\mathbf{q}_T) \\ \text{s.t.} \quad & \mathbf{q}_{t+1} = f(\mathbf{q}_t, \mathbf{u}_t), \quad 0 \leq t < T, \\ & \mathbb{P}[\mathcal{A}_t \cap \mathcal{O}_t = \emptyset] \geq 1 - \alpha, \quad 1 \leq t \leq T. \end{aligned} \quad (3)$$

Here safety is required at the *per-step* level—each step collision-free with probability at least $1 - \alpha$ —which is the guarantee our conformal construction certifies in the closed

¹Extending this to consider bidirectional interactions among the robot and dynamic agents is left as a future work.

loop (Theorems 1–2). The condition $\mathcal{A}(t) \cap \mathcal{O}(t) = \emptyset$ may be strengthened into

$$\inf_{x \in \mathcal{A}_t} D(x, \mathcal{O}_t) > \delta \quad (4)$$

where $\delta > 0$ is a user-specified margin between $\mathcal{A}_t$ and $\mathcal{O}_t$. Since $\mathcal{O}_t$ is generally non-convex, the exact verification of this condition is intractable even when $\mathcal{A}_t$ is a convex polytope (citation?). Therefore, a typical remedy for this is to approximate $\mathcal{O}_t$ or the free space $\mathcal{X} \setminus \mathcal{O}_t$ by a union of convex obstacles [26], [27]. Instead of inheriting this assumption, we simply assume that $\mathcal{A}(q)$ is circular:

$$\mathcal{A}(q) = \{x' \in \mathcal{X} : \|x' - x(q)\| \leq r_{\text{robot}}\},$$

in which case the constraint (4) can be simplified into

$$D(x_t, \mathcal{O}_t) \geq \delta + r_{\text{robot}}. \quad (5)$$

This can be simply checked by querying the distance function $D(\cdot, \mathcal{O}_t)$ once at x_t . The condition (3) is now translated into

$$\mathbb{P}[D(x_t, \mathcal{O}_t) \geq \delta + r_{\text{robot}}] \geq 1 - \alpha, \quad 1 \leq t \leq T. \quad (6)$$

To solve Problem 1, the policy π should incorporate information collected up to t : $\mathcal{H}_t := (\mathcal{O}_0, \dots, \mathcal{O}_t)$ to obtain future safety while maintaining overall deployment efficiency. A feasible approach that enables this is to define $\pi(\mathcal{H}_t)$ to be the solution of the following MPC problem:

$$\begin{aligned} \min_{\mathbf{q}, \mathbf{u}} \quad & \mathcal{L}(\mathbf{q}, \mathbf{u}, \mathbf{x}^{\text{ref}}) \\ \text{s.t.} \quad & q_{t|t} = q_t, \\ & q_{t+i+1|t} = f(q_{t+i|t}, u_{t+i|t}), \quad 0 \leq i < N, \\ & D(x_{t+i|t}, \mathcal{O}_{t+i|t}) \geq \delta + r_{\text{robot}}, \quad 1 \leq i \leq N. \end{aligned} \quad (7)$$

where $(x_{t+1|t}, \dots, x_{t+N|t})$ is a planned path in $\mathcal{X}$ at t based on x_t and $\mathcal{H}_t$, and $(\mathcal{O}_{t+1|t}, \dots, \mathcal{O}_{t+N|t})$ are *predicted* occupancies at $t+1, \dots, t+N$ from $\mathcal{H}_t$. The function $\mathcal{L}$ is an objective function of the MPC problem defined as

$$\mathcal{L}(\mathbf{q}, \mathbf{u}, \mathbf{x}^{\text{ref}}) := \sum_{i=0}^{N-1} \ell(q_{t+i|t}, u_{t+i|t}, x_{t+i|t}^{\text{ref}}) + \ell(q_{t+N|t}, x_{t+N|t}^{\text{ref}})$$

In this paper, we adopt *frontier method* [28] that pursues both information gathering and efficiency. On the prediction side, several different approaches are available. One common and mature approach is to estimate the kinematic states of dynamic agents through detection and tracking dynamic agents from $\mathcal{H}_t$, then deploy motion prediction models to predict their future motions, which can be back-projected into $\mathcal{X}$ to infer $(\mathcal{O}_{t+1|t}, \dots, \mathcal{O}_{t+N|t})$. (reference SOTA forecasting models to be added) Another approach is to perform prediction in an end-to-end fashion, i.e., deploy a model that consumes raw lidar scans and predict their future shapes. In this paper, we adhere to the former approach. (supportive arguments to be added) Since the prediction results are erroneous, the associated distance functions may falsely predict the true occupancies of the environment. Abbreviating $D_{t+i|t}(x) := D(x, \mathcal{O}_{t+i|t})$, we define the *score function* as

$$S_{t+i|t}(x) := D_{t+i|t}(x) - D_{t+i}(x), \quad x \in \mathcal{X}. \quad (8)$$

Intuitively, $S_{t+i|t}(x) > 0$ if the distance from x to $\mathcal{O}_{t+i}$ is closer than predicted. Hence, x with $S_{t+i|t}(x) > 0$ may be unsafe. This implies that correcting the underestimation is essential to ensure the constraint (6).

V. FUNCTIONAL CONFORMAL PREDICTION FOR PROVABLE DISTANCE FUNCTION COVERAGE

A. Why Functional Conformal Prediction?

In contrast to prior conformal-prediction approaches in robotics, where a single scalar score function is typically considered, the error signal $S_{t+i|t}$ here is *functional*: each score is a function $S_{t+i|t}: \mathcal{X} \rightarrow \mathbb{R}$, an element of $L^2(\mathcal{X})$, rather than a scalar. This motivates a CP framework tailored to such functional objects, namely *functional conformal prediction (FCP)* [23]. The practical value of FCP rests on two empirical properties of the residual field, which we verify on the ETH-UCY benchmark. First, the field is approximately *time-invariant*: estimated from two temporally disjoint halves of the data, the per-cell residual statistics agree closely across the split (Fig. 3; cell-mean correlation $r = 0.71$ – 0.91 across eth, univ, and hotel), so $S_{t+i|t} \stackrel{d}{\approx} S_{t'+i|t'}$ for $t \neq t'$. Second, the field is *low-rank*: a few functional principal components already capture most of its variance (Fig. 4; 5–7 components reach 90% on most scenes), so it admits the compact expansion

$$S_{t+i|t}(x) \approx \bar{S}_i(x) + \sum_{k=1}^r \xi_t^{(k)} \psi_k(x), \quad r \ll \infty, \quad (9)$$

with FPCA modes $\{\psi_k\}_{k=1}^r \subset L^2(\mathcal{X})$ that are stable across t . The uncertainty is thus spatially structured and compressible, even though it is not reducible to a simple geometric cue such as path curvature or visitation density. These two properties—which we quantify on the ETH-UCY benchmark in Section VIII-B1 (Figs. 3 and 4)—are exactly what make FCP practical: the modes $\{\psi_k\}$ and the resulting envelope $U_{t+i|t}: \mathcal{X} \rightarrow \mathbb{R}_{\geq 0}$ are estimated *once, offline*, while online the controller merely evaluates $U_{t+i|t}(x)$ pointwise along its rollouts, keeping the per-step cost low. To be self-contained, we briefly explain how FCP is baked into our setting so that the learning errors are corrected for safety.

B. Sketch of Core Pipeline

Our goal is to construct a high-probability upper bound $U_{t+i|t}$ for $S_{t+i|t}$:

$$\mathbb{P}[S_{t+i|t}(x) \leq U_{t+i|t}(x) \text{ for all } x \in \mathcal{X}] \geq 1 - \alpha, \quad (10)$$

Once such $U_{t+i|t}$ is identified, this can be used to derive a probabilistic lower bound of S_{t+i} as follows:

$$\mathbb{P}[D_{t+i}(x) \geq \underbrace{D_{t+i|t}(x) - U_{t+i|t}(x)}_{:= L_{t+i|t}(x)} \text{ for all } x \in \mathcal{X}] \geq 1 - \alpha$$

Then, we replace the constraint (7) by the following

$$L_{t+i|t}(x_{t+i|t}) \geq \delta + r_{\text{robot}}, \quad 1 \leq i \leq N. \quad (11)$$

However, directly computing such a bound in the infinite-dimensional Hilbert space $\mathcal{H}$ is generally intractable and often

unnecessary. Instead, we leverage the functional conformal prediction framework [23] and compute a bound on a finite-dimensional projection of the score function $S_{t+i|t}$. Specifically, we adopt a *divide-and-conquer* strategy to obtain a legitimate bound $U_{t+i|t}$: Consider the decomposition of $S_{t+i|t}$

$$S_{t+i|t} = \check{S}_{t+i|t} + \underbrace{(S_{t+i|t} - \check{S}_{t+i|t})}_{:= R_{t+i|t}}. \quad (12)$$

An upper bound for $\check{U}_{t+i|t}$ for $\check{S}_{t+i|t}$, a *finite-dimensional* projection of $S_{t+i|t}$ whose construction will be discussed later, will be obtained via FCP. The projection residual $R_{t+i|t}$, which is negligible compared to $\check{S}_{t+i|t}$, will be bounded uniformly from above: $\|R_{t+i|t}\|_{\mathcal{X}} \leq \varepsilon_i$, which can be straightforwardly done by applying standard CP. These bounds will finally be combined to comprise the desired bound of $S_{t+i|t}$:

$$U_{t+i|t}(x) := \check{U}_{t+i|t}(x) + \varepsilon_i. \quad (13)$$

C. FCP-based Upper Bound

The most intricate part of the aforementioned pipeline is construction of the finite-dimensional projection $\check{S}_{t+i|t}$ of $S_{t+i|t}$ and its high-confidence upper bound $\check{U}_{t+i|t}$. To begin with, we define a p -dimensional subspace $\mathcal{V} \subset \mathcal{H}$ and let $\check{S}_{t+i|t} := \Pi_{\mathcal{V}}(S_{t+i|t})$, where $\Pi_{\mathcal{V}} : \mathcal{H} \rightarrow \mathcal{V}$ denotes the projection operator. Then, it remains to find a confidence set $\mathcal{C}_{t+i|t}$ that satisfies a distribution-free coverage guarantee of the form

$$\mathbb{P}[\check{S}_{t+i|t} \in \mathcal{C}_{t+i|t}] \geq 1 - \alpha, \quad (14)$$

Here, $\mathcal{C}_{t+i|t} \subset \mathcal{V}$ is a conformal prediction set defined on the projected space. In the following sections, we describe a specific choice of $\mathcal{V}$ and how $\check{U}_{t+i|t}$ is derived from the confidence set $\mathcal{C}_{t+i|t}$ following [23].

1) *Data-driven basis functions*: In our setting, the projection subspace $\mathcal{V}_i \subset \mathcal{H}$ is learned from data at each horizon index $i = 1, \dots, N$ via functional principal components. Concretely, let $\{S_{t+i|t}^{(n)}\}_{n=1}^N$ denote a collection of score functions at step i . We choose an orthonormal basis $\{\varphi_{i,j}\}_{j=1}^{p_i}$ spanning $\mathcal{V}_i$ as p_i leading principal components of this residual sample, in the standard L^2 sense.

For any $S \in \mathcal{H}$, define its coefficient vector in this basis by

$$\xi_i(S) := (\langle S, \varphi_{i,1} \rangle, \dots, \langle S, \varphi_{i,p_i} \rangle)^\top \in \mathbb{R}^{p_i}, \quad (15)$$

so that the projection admits the expansion

$$(\Pi_{\mathcal{V}} S)(x) = \sum_{j=1}^{p_i} \xi_{i,j}(S) \varphi_{i,j}(x).$$

The dimension p_i of $\mathcal{V}_i$ may be fixed or selected as a function of the sample size and approximation accuracy; we treat this choice as a modeling parameter.

a) Inductive conformal prediction in coefficient space:

We adopt the inductive conformal predictor (ICP) framework [23] to conformalize a high-density region in coefficient space. We split the coefficient samples $\{\xi^{(n)}\}_{n=1}^N$ into a training subset and a calibration subset. A parametric conformity score $g(\xi)$ is fitted on the training subset, and a threshold is chosen using the empirical quantile on the calibration subset.

b) *Gaussian mixture model and ellipsoidal coefficient regions*: We model the distribution of the projected coefficients $\xi \in \mathbb{R}^p$ using a K -component Gaussian mixture model (GMM). Specifically, we fit the mixture weights $\hat{\pi}_k$, means $\hat{\mu}_k$, and covariance matrices $\hat{\Sigma}_k$ to the training coefficients and obtain the estimated density

$$\hat{f}(\xi) = \sum_{k=1}^K \hat{\pi}_k \varphi(\xi; \hat{\mu}_k, \hat{\Sigma}_k),$$

where $\varphi(\cdot; \mu, \Sigma)$ denotes the multivariate Gaussian density.

c) *Conformity score*: Following the construction in Eqs. (8)–(9) of [23], we define a max-component pseudo-density conformity score based on the fitted GMM parameters:

$$g(\xi) := \max_{1 \leq k \leq K} \hat{\pi}_k \varphi(\xi; \hat{\mu}_k, \hat{\Sigma}_k).$$

This score assigns higher conformity to coefficients lying in high-density regions of at least one mixture component, thereby yielding ellipsoidal conformal regions in the coefficient space.

Let $\{\xi^{(n)}\}_{n \in \text{cal}}$ be the coefficients associated with the calibration set and define

$$\lambda_i := \text{Quantile}_{1-\alpha/2}(\{g(\xi^{(n)})\}_{n \in \text{cal}}).$$

Then the ICP prediction set in coefficient space is

$$\begin{aligned} \mathcal{T}_{t+i|t} &:= \{\xi \in \mathbb{R}^{p_i} : g(\xi) \geq \lambda_i\}, \\ \mathbb{P}[\xi_{t+i|t} \in \mathcal{T}_{t+i|t}] &\geq 1 - \alpha/2. \end{aligned} \quad (16)$$

Based on the confidence set $\mathcal{T}_{t+i|t}$, which lies in the coefficient space $\mathbb{R}^{p_i}$, we derive a high-confidence upper bound of $\check{S}_{t+i|t}$ as follows:

$$\check{U}_{t+i|t}(x) := \sup_{\xi \in \mathcal{T}_{t+i|t}} \xi^\top \varphi_i(x), \quad x \in \mathcal{X}.$$

In particular, it immediately follows from $\check{S}_{t+i|t}(x) = \xi_{t+i|t}^\top \varphi_i(x)$ and (16) that

$$\mathbb{P}[\check{S}_{t+i|t}(x) \leq \check{U}_{t+i|t}(x) \text{ for all } x \in \mathcal{X}] \geq 1 - \frac{\alpha}{2}. \quad (17)$$

It turns out that $\check{U}_{t+i|t}$ admits a closed-form expression

$$\check{U}_{t+i|t}(x) = \max_{1 \leq k \leq K} \left\{ \hat{\mu}_k^\top \varphi_i(x) + r_{k,i} \left(\varphi_i(x)^\top \hat{\Sigma}_k \varphi_i(x) \right)^{\frac{1}{2}} \right\}$$

The detailed derivation of the expression is given in Appendix.

D. Bounding Projection Residuals.

To translate coefficient-space uncertainty back to the original discretized function, we account for the projection residual in sup norm. Define the reconstruction error for sample $y^{(n)}$ by

$$e^{(n)} := \|y^{(n)} - \Pi_{\mathcal{V}}(y^{(n)})\|_{\mathcal{X}}.$$

Using the calibration set split, we set the projection residual to the conformal quantile

$$\varepsilon_i := \text{Quantile}_{1-\alpha/2}(\{e^{(n)}\}_{n \in \text{cal}}), \quad (18)$$

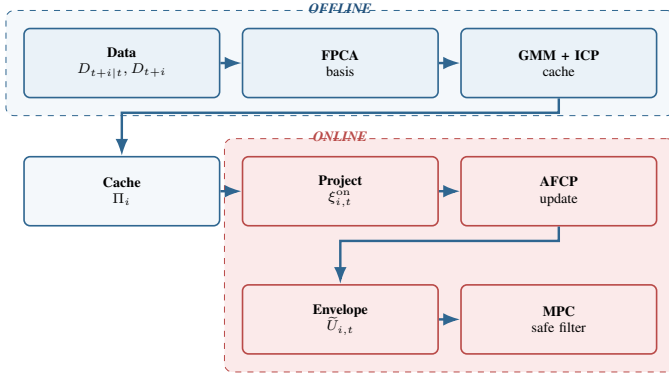


Fig. 1. Overview of the proposed functional CP-informed MPC. **Offline** (blue): predicted and realized signed-distance fields are reduced by FPCA and a GMM-based inductive conformal procedure into a cached coefficient-envelope tuple Π_i . **Online** (red): the realized residual is projected onto the cached basis, the coefficient upper bound is adapted by the AFCP rule (29), and the resulting envelope $\tilde{U}_{i,t}$ tightens the sampled-MPC safety filter. In the open 3D setting the online branch is unnecessary—ample free space makes the offline hard envelope sufficient—whereas the dense 2D benchmarks additionally exploit the online adaptation.

which yields

$$\mathbb{P} [|R_{t+i|t}(x)| \leq \varepsilon_i \text{ for all } x \in \mathcal{X}] \geq 1 - \alpha/2. \quad (19)$$

Combining (12), (13), (17), and (19), we finally get the desired coverage guarantee (10).

From computational side, the projection coefficients $\xi_{i,j}(S) = \langle S, \varphi_{i,j} \rangle$ generally require integration over $\mathcal{X}$, which is expensive when S is only available through costly distance-transform evaluations on a fine grid. Therefore, in implementation we approximate the projection (or its pointwise evaluations) using only finitely many grid points $\tilde{\mathcal{X}} \subset \mathcal{X}$.

VI. FCP-INFORMED MPC

We now incorporate the conformalized distance lower bound into a receding-horizon motion planner, following the general safe-planning template of [1] but adapting it to our *fieldwise* uncertainty representation. The central idea is to conformalize the safety bound over the *entire field* offline, rather than point by point online. Since the conformalized bound lives in a compact functional (coefficient) form, the expensive computation—fitting the basis and an accurate *initial* bound from a large calibration set through a GMM-based functional conformal procedure—is carried out once, offline, and cached. Online, the planner no longer recomputes per-point distances; it simply consults this precomputed field, projecting the realized residuals onto the cached basis and applying a lightweight update. Two benefits follow: the per-step cost stays well within real-time budgets, and the planner is handed a continuous, queryable safety field—directly suited to scoring the many candidate trajectories of an MPC rollout. The same update also absorbs the distribution shift that arises once the deployed error distribution drifts from the offline data, correcting it within the low-dimensional coefficient space rather than re-fitting the entire field (Section VI-C). Figure 1 summarizes the resulting pipeline.

A. MPC Formulation.

Incorporating the conformalized constraint (11) into (7) leads to the following MPC formulation:

$$\begin{aligned} \min_{\mathbf{q}, \mathbf{u}} \quad & \mathcal{L}(\mathbf{q}, \mathbf{u}, \mathbf{x}^{\text{ref}}) \\ \text{s.t.} \quad & \mathbf{q}_{t|t} = \mathbf{q}_t, \\ & \mathbf{q}_{t+i+1|t} = f(\mathbf{q}_{t+i|t}, \mathbf{u}_{t+i|t}), \quad 0 \leq i < N, \\ & \mathbf{L}_{t+i|t}(\mathbf{x}_{t+i|t}) \geq \delta + r_{\text{robot}}, \quad 1 \leq i \leq N. \end{aligned} \quad (20)$$

Rather than solving this nonlinear program exactly, a sampling-based MPC, such as model predictive path integral (MPPI) [29], can be used as a affordable implementation.

Hard Constraints vs Soft Penalization. While theoretically favorable, the conformalized constraints (20) may still produce extremely small feasible sets when i is large. This can be mitigated by modeling them implicitly as *soft constraints* [30], [31]: in place of enforcing the hard constraints, the constraints are incorporated into the objective function of the MPC problem as an extra graded term that penalizes the violation of the constraints. Specifically, we define

$$g_{t+i|t}(x) := \delta + r_{\text{robot}} - \mathbf{L}_{t+i|t}(x), \quad (21)$$

so that the constraint (20) is identical to $g_{t+i|t}(x) \leq 0$. Then, the soft version of the MPC is stated as follows:

$$\begin{aligned} \min_{\mathbf{q}, \mathbf{u}} \quad & \mathcal{L}_w(\mathbf{q}, \mathbf{u}, \mathbf{x}^{\text{ref}}) \\ \text{s.t.} \quad & \mathbf{q}_{t|t} = \mathbf{q}_t, \\ & \mathbf{q}_{t+i+1|t} = f(\mathbf{q}_{t+i|t}, \mathbf{u}_{t+i|t}), \quad 0 \leq i < N. \end{aligned} \quad (22)$$

where $\mathcal{L}_w$ is defined as

$$\mathcal{L}_w(\mathbf{q}, \mathbf{u}, \mathbf{x}^{\text{ref}}) = \mathcal{L}(\mathbf{q}, \mathbf{u}, \mathbf{x}^{\text{ref}}) + w \sum_{i=1}^N [g_{t+i|t}(\mathbf{x}_{t+i|t})]_+^2$$

for a user-defined constant $w > 0$. The penalty term discourages incursions into the conformalized unsafe region ($g_{t+i|t} > 0$). The weight $w > 0$ is fixed rather than adapted, thus (22) is a quadratic-penalty relaxation [32] of its hard counterpart (20): its minimizer approaches the hard-constrained solution as $w \rightarrow \infty$. For finite w , it stays well-defined even where the hard feasible set is empty, at the cost of the strict closed-loop guarantee. We compare both formulations in our 2D pedestrian experiments.

B. Stage 1: Offline Initialization

Dataset. During the offline stage, we assume an access to n_{cal} independent snapshots of the environment:

$$\left\{ \mathcal{O}_t^{(j)} : 0 \leq t \leq T, 1 \leq j \leq n_{\text{cal}} \right\}$$

from which we obtain the distance dataset for i -th planning step:

$$\mathcal{D}_{\text{cal},i} := \left\{ S_{t+i|t}^{(j)} : 0 \leq t \leq T-i, 1 \leq j \leq n_{\text{cal}} \right\}.$$

Then, we apply FPCA to obtain an orthonormal basis

$$\{\varphi_{i,1}, \dots, \varphi_{i,p_i}\},$$

This in turn produces the coefficient vector of each $S \in \mathcal{D}_{\text{cal},i}$ according to (15).

The key design principle is to separate the field-level *statistical initialization*, computed once offline, from the *lightweight adaptation* performed online. Offline, we exploit a large calibration set and a flexible GMM to construct the most accurate initial coefficient upper bound possible; this cost is paid only once, before deployment. Online, the GMM is no longer evaluated—the controller adapts the cached upper bound directly from streaming residual observations and a prescribed target violation level, so the per-step cost stays small.

a) *Time-to-go indexing.*: For each horizon index $i \in \{1, \dots, N\}$, we define the residual field

$$S_i(x) \equiv S_{t+i|t}(x) = D_{t+i|t}(x) - D_{t+i}(x), \quad x \in \mathcal{X}. \quad (23)$$

We assume that the distribution of residual fields depends primarily on the time-to-go index i , which allows the offline initialization to be performed once per horizon step and reused online.

b) *Offline CP via FPCA and GMM.*: For each i , we collect a calibration set

$$\mathcal{D}_{\text{cal},i} = \{S_i^{(1)}, \dots, S_i^{(n_{\text{cal}})}\}.$$

We then apply FPCA to obtain an orthonormal basis

$$\{\varphi_{i,1}, \dots, \varphi_{i,p_i}\},$$

and represent each residual field by its coefficient vector

$$\xi_i(S) = (\langle S, \varphi_{i,1} \rangle, \dots, \langle S, \varphi_{i,p_i} \rangle)^\top \in \mathbb{R}^{p_i}. \quad (24)$$

Using the coefficient dataset $\{\xi_i(S_i^{(j)})\}_{j=1}^{n_{\text{cal}}}$, we fit a Gaussian mixture model and apply the coefficient-space conformal construction of Section V-C1b. This yields the mixture parameters $\{\hat{\pi}_{i,k}, \hat{\mu}_{i,k}, \hat{\Sigma}_{i,k}\}_{k=1}^K$, the conformal level λ_i (and hence the per-component radii $r_{i,k} \equiv r_{i,k}(\lambda_i)$ of (39)), and the projection residual ε_i . The initial offline envelope is then the closed-form support function of the resulting union of ellipsoids,

$$U_i^{(0)}(x) = \varepsilon_i + \max_{1 \leq k \leq K} \left\{ \hat{\mu}_{i,k}^\top \varphi_i(x) + r_{i,k} (\varphi_i(x)^\top \hat{\Sigma}_{i,k} \varphi_i(x))^{1/2} \right\}, \quad (25)$$

where $\varphi_i(x) \in \mathbb{R}^{p_i}$ denotes the vector of basis evaluations at x . (The support function automatically takes the worst-case direction over each ellipsoid, so it is a valid upper bound regardless of the sign of the basis functions; a single per-coordinate vector would not be.)

c) *Cached offline tuple.*: For each horizon index i , we cache the offline tuple

$$\Pi_i := \left(\{\varphi_{i,m}\}_{m=1}^{p_i}, \{\hat{\pi}_{i,k}, \hat{\mu}_{i,k}, \hat{\Sigma}_{i,k}\}_{k=1}^K, \lambda_i, \varepsilon_i \right), \quad (26)$$

and denote by

$$\mathbf{\Pi} := \{\Pi_i\}_{i=1}^N \quad (27)$$

the full collection of horizon-indexed tuples cached by the controller. Crucially, after deployment the online controller only *evaluates* the closed-form envelope (25) from the cached parameters and never *re-fits* the GMM. In practice the two grids $\hat{\mu}_{i,k}^\top \varphi_i(\cdot)$ and $(\varphi_i(\cdot)^\top \hat{\Sigma}_{i,k} \varphi_i(\cdot))^{1/2}$ are precomputed once, so an envelope update reduces to recombining them with the (scalar-)adapted radii.

C. Stage 2: Online Adaptation

Once the robot starts moving, the cached envelope is adapted online through a single scalar per horizon index, in the spirit of adaptive conformal inference [24]. Because the offline envelope (25) is parameterized by the calibrated radii $r_{i,k}$, we adapt a scalar multiplier $c_i^{(t)}$ on those radii rather than a high-dimensional coefficient vector: this keeps the band a valid LRW support function at all times and yields a clean coverage event. At time t , once a realized residual field $S_{i,t}^{\text{on}}$ matures for horizon index i , we form the *functional* violation indicator

$$\text{err}_i^{(t)} = \mathbb{I}[\exists x \in \bar{\mathcal{X}} : S_{i,t}^{\text{on}}(x) > \tilde{U}_{i,t}(x)], \quad (28)$$

evaluated only on the discretization grid $\bar{\mathcal{X}}$ of Section V-B (cost $O(Kp_i|\bar{\mathcal{X}}|)$), no full-field reconstruction and no GMM re-fit). The scalar is then updated by the ACI rule

$$c_i^{(t+1)} = \left[c_i^{(t)} + \gamma (\text{err}_i^{(t)} - \alpha_{\text{target}}) \right]_+, \quad (29)$$

so the envelope inflates when the realized miscoverage exceeds α_{target} and contracts otherwise, driving the long-run miscoverage to α_{target} . The corresponding online envelope simply rescales the cached radii,

$$\tilde{U}_{i,t}(x) = \varepsilon_i + \max_{1 \leq k \leq K} \left\{ \hat{\mu}_{i,k}^\top \varphi_i(x) + c_i^{(t)} r_{i,k} (\varphi_i(x)^\top \hat{\Sigma}_{i,k} \varphi_i(x))^{1/2} \right\}. \quad (30)$$

Thus offline GMM fitting only initializes the cached parameters, while online adaptation moves a single scalar $c_i^{(t)}$ (with $c_i^{(0)} = 1$), recomputed against the precomputed grids of (26) without re-evaluating the GMM.

a) *Optional grid realization for fast online evaluation.*:

Although the primary cached objects are the tuples in (27), the envelope (30) may also be evaluated on a fixed workspace grid $\mathcal{G}$ and stored as a tensor for fast pointwise lookup. This grid realization is purely an implementation-level acceleration.

b) *Pointwise online evaluation along rollouts.*: For each candidate trajectory and each horizon step, the controller evaluates

$$D_{t+i|t}(p_{t+i|t}^{(k)}) = \max \left\{ D_{t+i|t}(p_{t+i|t}^{(k)}) - \tilde{U}_{i,t}(p_{t+i|t}^{(k)}), 0 \right\},$$

and accepts the trajectory only if $x_{t+i|t}^{(k)} \in \mathcal{S}_{t+i|t}^{\text{cp}}$ (??) holds at every step.

Note on Computational Implications. This decomposition has two important consequences. First, the computationally expensive statistical modeling is performed only once offline, where large-scale data and a flexible GMM can be used to obtain an accurate initialization. Second, the online controller avoids all GMM computations and updates only the low-dimensional coefficient upper bound $\hat{\xi}_i^{(t)}$, which substantially reduces online overhead and better matches real-time operation.

Algorithm 1 summarizes the resulting procedure.

VII. SAFETY ANALYSIS AND ASYMPTOTIC GUARANTEES

This section establishes the safety guarantees of the proposed framework. A key advantage of the functional formulation is that pointwise safety holds for *any* planned trajectory

Algorithm 1 Functional CP-informed MPC with offline GMM initialization and online coefficient-space adaptive update

- 1: **Input:** Calibration set $\{\mathcal{D}_{\text{cal},i}\}_{i=1}^N$, safety radius r_{safe} , tracking margin ϵ , target violation level α_{target} , workspace grid $\mathcal{G}$.
 - 2: **Offline initialization:**
 - 3: **for** each horizon index $i = 1, \dots, N$ **do**
 - 4: Compute residual-field samples $S_i^{(j)}(x) = D_{t+i|t}^{(j)}(x) - D_{t+i}^{(j)}(x)$;
 - 5: Apply FPCA and extract basis $\{\varphi_{i,m}\}_{m=1}^{p_i}$;
 - 6: Project all residual fields to obtain coefficient vectors $\xi_i^{(j)}$;
 - 7: Fit a GMM in coefficient space and compute the initial coefficient upper bound $\hat{\xi}_i^{(0)}$;
 - 8: Compute the projection residual ε_i ;
 - 9: Cache the tuple $\Pi_i = (\{\varphi_{i,m}\}_{m=1}^{p_i}, \hat{\xi}_i^{(0)}, \varepsilon_i)$;
 - 10: **end for**
 - 11: Retain only the cached tuples $\{\Pi_i\}_{i=1}^N$; the GMM is not used online.
 - 12: **Online MPC loop** (online adaptation optional):
 - 13: **for** $t = 0, 1, 2, \dots$ **do**
 - 14: Observe current robot state z_t and obtain obstacle predictions $\{\hat{\mathcal{O}}_{t+i|t}\}_{i=1}^N$;
 - 15: **for** each horizon index i for which an online residual field $S_{i,t}^{\text{on}}$ becomes available **do**
 - 16: Project $S_{i,t}^{\text{on}}$ onto the cached basis and compute $\xi_{i,t}^{\text{on}}$;
 - 17: Update the coefficient upper bound $\hat{\xi}_i^{(t+1)}$ (29);
 - 18: **end for**
 - 19: Sample candidate input sequences and roll out candidate trajectories;
 - 20: Evaluate $\tilde{U}_{i,t}(x) = \varepsilon_i + \hat{\xi}_i^{(t)} \top \varphi_i(x)$ along rollouts;
 - 21: Filter infeasible trajectories using $\underline{D}_{t+i|t}(x) \geq r_{\text{safe}} + \epsilon$;
 - 22: Select the feasible candidate minimizing the MPC objective (20);
 - 23: Apply the first control input and advance to the next planning step;
 - 24: **end for**
-

satisfying the tightened conformal constraint, independent of how the control space is sampled—precisely because the bound is certified over the entire field rather than along a single path.

We first present the base safety theorem, assuming the functional envelope covers the true score; we then extend it to a robust guarantee for dynamic environments through the online adaptive update.

A. Base Safety Guarantee via Functional Coverage

The following theorem demonstrates that if the spatial envelope asymptotically covers the true score function, the MPC framework guarantees the safety of the closed-loop system.

Theorem 1 (Asymptotic Safety Guarantee). *Suppose that the CP-MPC problem (20) is feasible for all $t \geq 0$. Let p_{t+1} be the*

actual robot position at time $t + 1$, and suppose the tracking error satisfies

$$\|p_{t+1} - p_{t+1|t}^*\| \leq \epsilon.$$

Further suppose that the online envelope $\tilde{U}_{1,t}$ satisfies the asymptotic coverage property

$$\lim_{T \rightarrow \infty} \frac{1}{T} \sum_{t=0}^{T-1} \mathbb{I}[S_{t+1|t}(x) \leq \tilde{U}_{1,t}(x) \quad \forall x \in \mathcal{X}] = 1 - \alpha \quad \text{w.p. 1.}$$

Then the closed-loop system is asymptotically $(1 - \alpha)$ -safe:

$$\liminf_{T \rightarrow \infty} \frac{1}{T} \sum_{t=0}^{T-1} \mathbb{I}[D_{t+1}(p_{t+1}) \geq r_{\text{safe}}] \geq 1 - \alpha, \quad \text{w.p. 1.} \quad (31)$$

Proof. For each t , define the event

$$\mathcal{E}_{t+1} := \{S_{t+1|t}(x) \leq \tilde{U}_{1,t}(x) \quad \forall x \in \mathcal{X}\}.$$

On the event $\mathcal{E}_{t+1}$, the definition $S_{t+1|t} = D_{t+1|t} - D_{t+1}$ gives, for all $x \in \mathcal{X}$,

$$D_{t+1}(x) \geq D_{t+1|t}(x) - \tilde{U}_{1,t}(x).$$

Since the MPC problem is feasible, the applied plan lies in the tightened safe set (??), so by (??)

$$\underline{D}_{t+1|t}(p_{t+1|t}^*) \geq r_{\text{safe}} + \epsilon > 0. \quad (32)$$

Because the clipping in (??) is therefore inactive at $p_{t+1|t}^*$, we have $\underline{D}_{t+1|t}(p_{t+1|t}^*) = D_{t+1|t}(p_{t+1|t}^*) - \tilde{U}_{1,t}(p_{t+1|t}^*)$, and hence on $\mathcal{E}_{t+1}$

$$D_{t+1}(p_{t+1|t}^*) \geq \underline{D}_{t+1|t}(p_{t+1|t}^*) \geq r_{\text{safe}} + \epsilon. \quad (33)$$

Next, by Property ?? and the tracking assumption,

$$\begin{aligned} D_{t+1}(p_{t+1}) &\geq D_{t+1}(p_{t+1|t}^*) - \|p_{t+1} - p_{t+1|t}^*\| \\ &\geq D_{t+1}(p_{t+1|t}^*) - \epsilon. \end{aligned} \quad (34)$$

Combining (33) and (34), we obtain, on $\mathcal{E}_{t+1}$,

$$D_{t+1}(p_{t+1}) \geq (r_{\text{safe}} + \epsilon) - \epsilon = r_{\text{safe}}.$$

Hence,

$$\mathbb{I}[D_{t+1}(p_{t+1}) \geq r_{\text{safe}}] \geq \mathbb{I}[\mathcal{E}_{t+1}].$$

Averaging over $t = 0, \dots, T - 1$ and taking $\liminf$ yields

$$\begin{aligned} \liminf_{T \rightarrow \infty} \frac{1}{T} \sum_{t=0}^{T-1} \mathbb{I}[D_{t+1}(p_{t+1}) \geq r_{\text{safe}}] &\geq \lim_{T \rightarrow \infty} \frac{1}{T} \sum_{t=0}^{T-1} \mathbb{I}[\mathcal{E}_{t+1}] \\ &= 1 - \alpha \end{aligned}$$

with probability one, which proves (31). $\square$

B. Robust Guarantee under Online Adaptation

Theorem 1 establishes the foundational logic mapping functional coverage to pointwise safety. In offline CP settings, guaranteeing the exact equality of the coverage (i.e., $\lim_{T \rightarrow \infty} \dots = 1 - \alpha$) heavily relies on the exchangeability of the calibration and test data. However, in dynamic environments, unpredicted distribution shifts can invalidate this assumption.

To robustify our framework against such shifts, we drop the exchangeability assumption and employ the online Adaptive Functional Conformal Prediction (AFCP) method. By instantiating the adaptive conformal inference of Section III in the low-dimensional coefficient space (the online update of Section VI-C), the AFCP update rule drives the long-run coverage to a robust lower bound:

$$\liminf_{T \rightarrow \infty} \frac{1}{T} \sum_{t=0}^{T-1} \mathbb{I}[S_{t+1|t}(x) \leq \tilde{U}_{1,t}(x) \quad \forall x \in \mathcal{X}] \geq 1 - \alpha \quad (35)$$

with probability one.

Based on this robust property, the following theorem extends the safety guarantee to dynamic, non-exchangeable environments.

Theorem 2 (Asymptotic Safety with AFCP). *Consider a dynamic environment where data exchangeability does not hold. Suppose the feasibility and tracking conditions of Theorem 1 are satisfied. If the functional envelope $\tilde{U}_{1,t}$ is updated online via the AFCP rule satisfying (35), then the closed-loop system robustly maintains the asymptotic $(1 - \alpha)$ -safety:*

$$\liminf_{T \rightarrow \infty} \frac{1}{T} \sum_{t=0}^{T-1} \mathbb{I}[D_{t+1}(p_{t+1}) \geq r_{\text{safe}}] \geq 1 - \alpha, \quad \text{w.p. } 1. \quad (36)$$

Proof. Let $\mathcal{E}_{t+1} := \{S_{t+1|t}(x) \leq \tilde{U}_{1,t}(x) \quad \forall x \in \mathcal{X}\}$ be the functional coverage event. Following the exact same derivations as in the proof of Theorem 1—specifically regarding the tightened MPC constraint (32) and the Lipschitz continuity of the SDF (34)—we establish that for any step t , the functional coverage directly implies point-wise safety: $\mathbb{I}[D_{t+1}(p_{t+1}) \geq r_{\text{safe}}] \geq \mathbb{I}[\mathcal{E}_{t+1}]$.

Taking the average over T steps and applying the limit infimum on both sides, we obtain:

$$\liminf_{T \rightarrow \infty} \frac{1}{T} \sum_{t=0}^{T-1} \mathbb{I}[D_{t+1}(p_{t+1}) \geq r_{\text{safe}}] \geq \liminf_{T \rightarrow \infty} \frac{1}{T} \sum_{t=0}^{T-1} \mathbb{I}[\mathcal{E}_{t+1}].$$

By substituting the AFCP coverage guarantee (35) into the right-hand side, we immediately conclude:

$$\liminf_{T \rightarrow \infty} \frac{1}{T} \sum_{t=0}^{T-1} \mathbb{I}[D_{t+1}(p_{t+1}) \geq r_{\text{safe}}] \geq 1 - \alpha.$$

This completes the proof, verifying that the point-wise safety of the robot is inherently guaranteed by the online adaptation of the functional envelope. $\square$

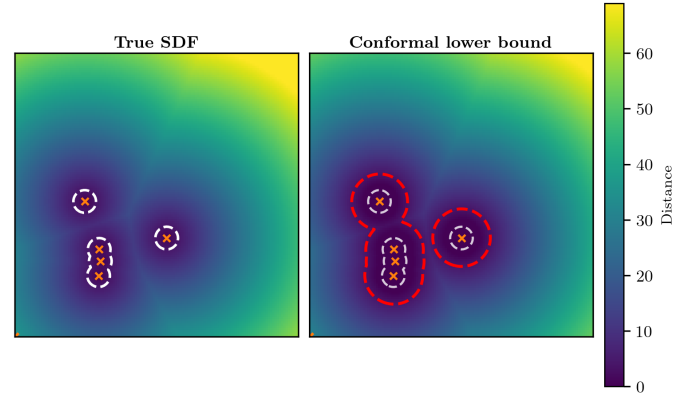


Fig. 2. 2D conformal SDF visualization at a representative time. **Left:** ground-truth signed distance field. **Right:** conformal lower-bound field. Dashed contours indicate the safety threshold r_{safe} .

VIII. EXPERIMENTAL SETUP

We evaluate the proposed *Conformal SDF* framework and its integration into a model predictive controller. Our experiments are designed to assess how functional conformal prediction of the signed distance field affects safety, feasibility, and efficiency in closed-loop motion planning under prediction uncertainty.

To this end, we conduct experiments at increasing levels of complexity: a 2D toy setting for visualizing conformalized distance fields, standard 2D pedestrian navigation benchmarks for quantitative evaluation, and a challenging 3D quadrotor navigation task for assessing closed-loop performance under dense, uncertain dynamics.

Across all settings, conformal prediction is applied to conformalize a lower bound of the predicted signed distance field, which is then enforced as a hard safety constraint within MPC.

a) CP: We collect a calibration set of length T_{cal} and compute residual SDF fields $R_{t+i|t}(x) = D_{t+i|t}(x) - D_{t+i}(x)$ over a 3D grid for each horizon step $i \in \{1, \dots, H\}$. The residual fields are projected onto a p -dimensional subspace using PCA and modeled via a K -component Gaussian mixture model (GMM) in coefficient space. We set the target miscoverage level to α and compute horizon-dependent upper envelopes $U_{t+i|t}(x)$, yielding the conformal lower-bound SDF defined in (??).

A. Conformal SDF Visualization at 2D

Figure 2 shows an illustrative 2D example where the conformal lower bound expands the unsafe region in areas of high uncertainty.

B. CP-Informed MPC in 2D Pedestrian Environments

Before evaluating the proposed framework in complex 3D scenarios, we first validate its effectiveness in standard 2D pedestrian navigation benchmarks. These experiments serve as a controlled testbed to examine how spatial conformal prediction affects safety, feasibility, and efficiency in a well-studied setting.

a) *2D Simulation Setup*: We evaluate all methods on all five ETH-UCY pedestrian datasets—eth, hotel, univ, zara1, and zara2. Pedestrian trajectories are taken from real-world recordings, and a mobile robot is tasked with navigating toward a fixed goal while avoiding dynamic pedestrians.

At each planning step, the robot observes recent pedestrian state histories and receives predicted future trajectories over a fixed planning horizon. A Trajectron++ [17] predictor is used as the base prediction model. This setup allows us to isolate the effect of *offline error quantification* independently of the prediction model itself.

1) *Spatial regularity of the residual field*: Before the closed-loop comparison, we verify on ETH-UCY the two properties that justify calibrating the envelope offline (Section V-A). Figure 3 tests *time-invariance*: estimating the per-cell residual statistics from two temporally disjoint halves of each scene yields closely agreeing fields (cell-mean correlation $r = 0.71$ – 0.91 across eth, univ, and hotel), so the field is a stable property of the scene rather than of a particular time. Figure 4 tests *low rank*: a functional PCA of the residual fields shows that 5–7 components already explain 90% of the variance on most scenes, and the leading eigenfunction is a smooth, scene-frame spatial mode. Together these confirm that the residual field is spatially structured and compressible without being reducible to a simple cue such as path curvature or visitation density—the premise that lets an offline-calibrated envelope transfer online at low cost.

2) *Robot dynamics*: Let the robot state be $z_t = (x_t, y_t, \theta_t) \in \mathbb{R}^2 \times S^1$ and the control input be $u_t = (v_t, \omega_t) \in \mathcal{U}$. We use the discrete-time unicycle model

$$z_{t+1} = f_r(z_t, u_t) = \begin{bmatrix} x_t + \Delta t v_t \cos \theta_t \\ y_t + \Delta t v_t \sin \theta_t \\ \theta_t + \Delta t \omega_t \end{bmatrix}.$$

The robot is modeled as a unicycle system with bounded linear and angular velocities. For a fair comparison, all methods share the same trajectory sampler, planning horizon, and cost structure, including goal-tracking and control-effort terms.

We compare four controllers spanning how prediction uncertainty is handled:

- *CC-MPC*: safety through soft penalty terms, without explicit uncertainty quantification;
- *ECP-MPC*: explicit online reasoning about prediction uncertainty;
- *ACP-MPC*: online adaptation of conformal bounds; and
- *FCP-MPC* (ours): offline functional conformal prediction of the signed-distance field.

For FCP-MPC we evaluate both a hard- and a soft-constraint formulation, ablating in each case the online coefficient adaptation of (29) to assess its effect on overconservatism at long horizons.

a) *Evaluation Metrics*: We report four closed-loop metrics (lower is better), averaged over the scenes of each dataset:

- *collision rate*: fraction of planning steps that violate the safety threshold;
- *infeasible rate*: fraction of steps for which no feasible trajectory is found;

- *time-to-goal*: number of planning steps to reach the goal; and
- *control time*: average per-step computation (ms).

b) *Hard vs. soft use of the conformal bound in 2D*:

In the 2D pedestrian benchmarks, we evaluate two variants of FCP-MPC that differ in how the conformal lower-bound signed distance field is incorporated.

Hard-CP enforces the conformal bound strictly as a hard feasibility constraint over the entire planning horizon. In dense 2D scenes, the offline functional envelope can become overly conservative as the horizon increases due to compounding prediction errors, which significantly shrinks the feasible set and leads to a high infeasible rate.

Soft-CP instead incorporates the conformal lower bound only as a soft safety-shaping penalty in the MPC objective: it keeps a hard feasibility filter for static obstacles but not for the predicted dynamic obstacles, ranking candidates by their conformal safety margin. Because the dynamic conformal bound no longer constrains the feasible set, the planner rarely exhausts its feasible candidates, trading the worst-case feasibility guarantee for consistently low infeasibility and shorter paths. Table I reports the main comparison against the baselines, while Table II isolates the effect of the online envelope adaptation within each variant.

C. CP-Informed MPC in 3D

We embed the conformal SDF into MPC by replacing the nominal distance constraint with the conformal lower bound:

$$\underline{D}_{t+i|t}(x_{t+i}) \geq r_{\text{safe}}, \quad i = 1, \dots, H. \quad (37)$$

a) *3D Experimental Setup*: We evaluate FCP-MPC in a challenging physics-based quadrotor navigation task implemented in PyBullet. The robot is modeled as a quadrotor with a low-level PID controller running at 240 Hz, while the high-level MPC planner operates at a slower planning period $\Delta t = 0.1$ s.

The environment contains $M = 280$ dynamic obstacles, modeled as kinematic agents with stochastic mode switching among multiple motion patterns, including constant-velocity, turning, wandering, and stop-and-go behaviors. Additive process noise is applied to obstacle dynamics, inducing increasing prediction uncertainty over the planning horizon. Obstacles are rendered as spheres within a bounded 3D workspace $x, y \in [-3, 7]$, $z \in [0, 8]$.

At each planning step the controller receives recent obstacle state histories and predicted obstacle trajectories over a horizon $H = 12$; oracle future trajectories are provided solely for evaluation and metric computation. A constant-velocity predictor is used as the base prediction model, and prediction noise is injected to emulate model mismatch.

Due to the higher-dimensional workspace and the availability of sufficient free space in 3D, we employ a *fully offline* functional conformal prediction in this setting. Unlike the 2D case, excessive conservativeness does not immediately block progress, allowing the conformalized worst-case safety envelopes to be enforced directly without additional online adaptation.

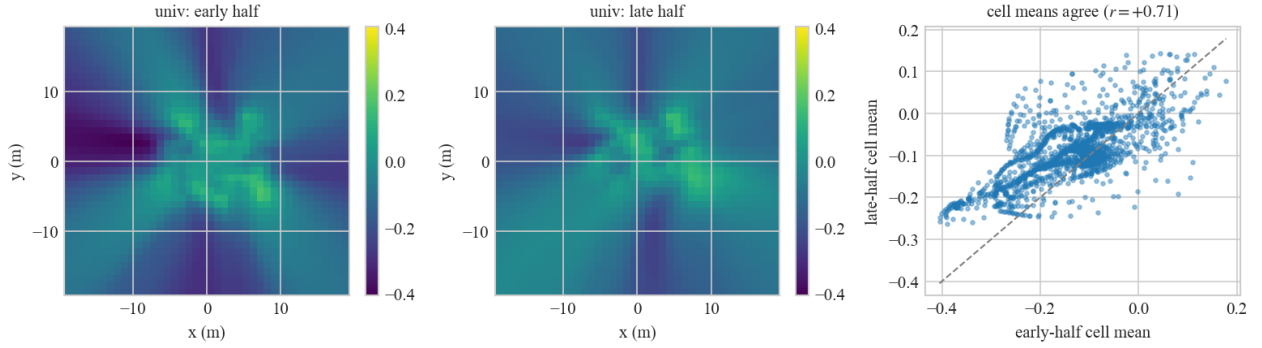


Fig. 3. The residual field is approximately *time-invariant*. Per-cell residual means estimated from temporally disjoint early (**left**) and late (**middle**) halves of `univ` agree closely (**right**; cell-mean correlation $r = +0.71$), indicating the field is a stable property of the scene rather than of a particular time — the premise that lets an offline-calibrated envelope transfer online.

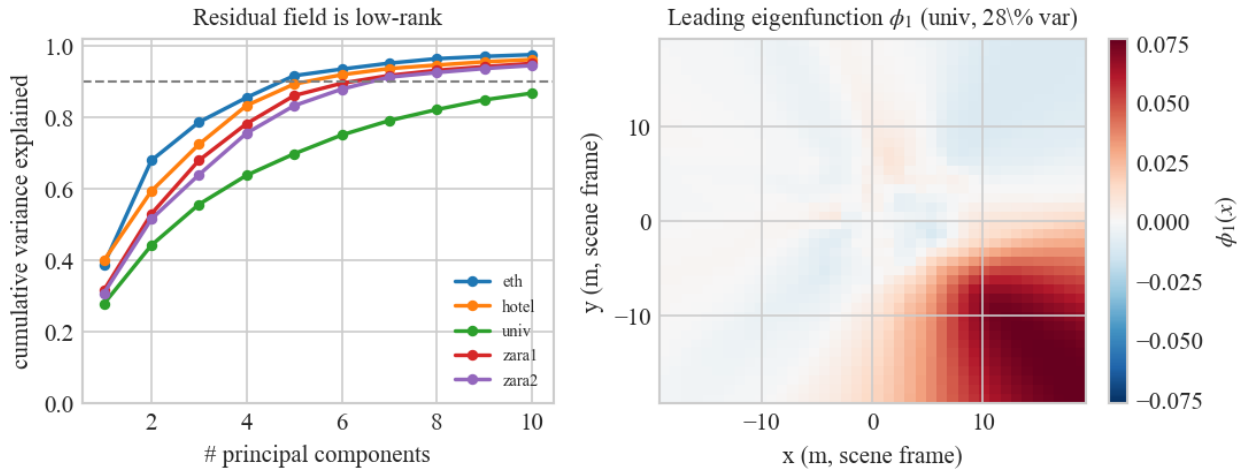


Fig. 4. The residual score field is *low-rank*. **Left**: cumulative variance explained by the leading functional principal components on each ETH-UCY scene; 5–7 components already reach 90% (dashed) on most scenes. **Right**: the leading eigenfunction ψ_1 (`univ`), a smooth scene-frame spatial mode. The low effective dimension is what lets the envelope be learned offline.

The planner samples endpoint candidates and generates smooth quintic polynomial trajectories in $\mathbb{R}^3$. Unsafe trajectories are filtered using the conformal lower-bound SDF constraint, and the minimum-cost feasible trajectory is selected. Each episode terminates once the robot reaches the goal within 0.25 m or a maximum step limit is reached.

b) Evaluation Metrics: We use the same closed-loop metrics as in the 2D study—collision rate, infeasible rate, time-to-goal, and per-step runtime—with collision now defined by the minimum robot–obstacle distance violating r_{safe} . All metrics are averaged over the evaluated seeds.

IX. RESULTS AND ANALYSIS

Qualitative rollouts indicate that FCP-MPC adapts conservativeness to spatially varying uncertainty: it becomes more conservative in crowded or uncertain regions while remaining less restrictive in open areas.

A. 2D Pedestrian Navigation

Table I reports quantitative results on the ETH-UCY pedestrian datasets, and Figure 5 shows representative closed-loop

trajectories of FCP-MPC and the baselines. Several consistent trends emerge across all environments.

CC-MPC has a zero infeasible rate by construction (soft penalty), but is slow—it times out in the most crowded scene (`univ`)—and its collisions, while below nominal planning, remain non-negligible in dense scenes. ECP-MPC’s explicit online uncertainty reasoning dominates its cost: its control time (42–78 ms) approaches or exceeds the planning period, inflating its infeasible rate and causing timeouts (`hotel`, `univ`, `zara2`), while its collision rate is only middling (e.g. 0.204 on `univ`). ACP-MPC is cheap and attains low collisions on some scenes (e.g. 0.000 on `univ`), but its scalar online adaptation leaves high infeasible rates and frequent timeouts in dense, dynamic scenes.

FCP-MPC exposes a clear safety–feasibility trade-off between its two deployments of the same offline field. The hard variant attains the lowest collision rate where free space allows (e.g. 0.023 on `eth`) but keeps a non-zero infeasible rate, as the worst-case envelope compounds over the horizon. The soft variant removes infeasibility on every scene and gives

the shortest paths (lowest steps-to-goal on all five scenes), at the cost of higher collisions in the densest scenes (*univ*, *zara2*). Both keep control time far below ECP-MPC (a few milliseconds vs. tens), because uncertainty is quantified offline and only a low-dimensional coefficient update runs online. Rather than dominating every metric, FCP-MPC offers a favorable safety–feasibility–efficiency trade-off: the same calibrated field can be enforced strictly (hard) or used to shape the cost (soft) as the scene demands.

a) Effect of online envelope adaptation: The offline functional envelope is conformalized on held-out data and can become overly conservative under the distribution shift of dense, dynamic scenes, occasionally eliminating all feasible trajectories after hard filtering. The online update (29) addresses this by adapting the coefficient upper bound $\hat{\xi}_i^{(t)}$ directly from realized residuals, without re-invoking the GMM. Table II isolates this effect. For the hard variant, online adaptation markedly reduces the infeasible rate—from 0.480 to 0.265 on *eth*, from 0.640 to 0.383 on *univ*, and from 0.454 to 0.262 on *zara1*—by relaxing the envelope where the offline bound was unnecessarily tight, while leaving collision rates essentially unchanged. For the soft variant, which is already feasible by construction, adaptation yields only marginal changes. Because the update operates on the low-dimensional coefficients evaluated on the discretization grid of Section V-B, it adds only a few milliseconds of online computation and stays well below the cost of ECP-MPC.

B. 3D Quadrotor Navigation

Table III reports closed-loop results at $M = 280$ dynamic obstacles and noisy predictions (prediction noise 0.20, obstacle process noise 0.22, oracle future noise 0.20), averaged over 17 seeds. The high-level planner runs at $\Delta t = 0.1$ s, so any per-step cost above 100 ms violates the real-time budget.

Neither uncertainty-reasoning baseline reaches the goal. ACP-MPC [2] inflates a scalar conformal radius per obstacle; this keeps its collision (0.032) and infeasible (0.130) rates low, but the per-step adaptation costs 217 ms and the quadrotor loses control authority before reaching the goal (crashed). ECP-MPC [5] is the slowest (306 ms/step) and the least safe (0.090 collision, 0.342 infeasible), and likewise crashes. CC-MPC does reach the goal (186.5 steps) thanks to its purely soft formulation (zero infeasible by construction), but at a higher collision rate (0.064) and a per-step cost (215 ms) that also exceeds the budget.

Both deployments of the *same* offline-conformalized field reach the goal. FCP-MPC (hard) enforces the bound as a hard feasibility filter and inherits the guarantee of Theorems 1–2; it reaches the goal (124 steps) at the lowest per-step cost of any method (51.5 ms—the only method within the 100 ms budget) and a low collision rate (0.044), though the worst-case envelope leaves a non-zero infeasible rate (0.269). FCP-MPC (soft) applies the same field as a penalty: it removes infeasibility entirely, attains the lowest collision rate (0.032, within the target $\alpha = 0.10$) and the fewest steps to goal (84.8); its per-step cost (160 ms) exceeds the budget but is still about 2× faster than ECP-MPC.

Table IV repeats the comparison at $N_{\text{obs}} = 50$: FCP-MPC again reaches the goal under both deployments with the lowest collision rates (0.012/0.015), while ACP- and ECP-MPC still crash; we discuss the density trend in Section IX-C. Figure 6 illustrates the conformal bound in 3D, and Figure 7 compares closed-loop trajectories across three seeds: FCP-MPC threads to the goal in every panel while the baselines stall or fall (red ×) early in the field.

a) Scalability: Figure 8 reports the mean per-step control time as the obstacle count grows from $N_{\text{obs}} = 10$ to 280. Because FCP-MPC checks safety by querying a *cached conformal field* rather than re-evaluating distances to every obstacle, its per-step cost is largely insensitive to the obstacle count and remains the lowest of all methods across the range (hard filter: 26 ms at $N_{\text{obs}} = 50$, 52 ms at 280). The baselines instead scale with the number of obstacles: CC- and ACP-MPC rise from tens of milliseconds to over 200 ms at $N_{\text{obs}} = 280$, and ECP-MPC—which also performs pointwise conformal reasoning across obstacles and horizon steps—is one to two orders of magnitude slower throughout and never meets the 100 ms budget. ACP-MPC, moreover, conformalizes only a single pointwise radius; FCP-MPC retains a calibrated safety *field* while being the most scalable method in dense 3D.

C. Discussion

a) Enforcing the safety bound in 2D: The 2D benchmarks isolate how the conformal lower bound should be *used* once it has been constructed. The hard variant attains the lowest collision rates but, because the worst-case envelope compounds over the horizon, can become overconservative and leave a non-zero infeasible rate; the soft variant removes this infeasibility by treating the bound as a penalty rather than a hard constraint, at the price of a higher collision rate in the densest scenes (Table I). The online coefficient adaptation of (29) directly counteracts the overconservatism of the hard variant, substantially lowering its infeasible rate while leaving collisions essentially unchanged (Table II). Both effects reuse the offline conformal prediction, so the added online cost is only a few milliseconds—far below the tens of milliseconds required by ECP-MPC’s explicit online uncertainty reasoning. These pedestrian experiments thus stress-test the fieldwise conformal prediction in a regime where free space is scarce, complementing the open 3D study below.

b) Separating conformal prediction from real-time planning in 3D: The three behaviors in Table III follow from *where* each controller spends its uncertainty budget. ECP-MPC reasons about uncertainty online and pays for it: its per-step latency far exceeds the 100 ms budget, so the quadrotor loses control authority and crashes before reaching the goal. ACP-MPC’s pointwise inflation is cheaper per query but still re-checks every obstacle, and it too crashes without reaching the goal. CC-MPC reaches the goal but, lacking any uncertainty model, is both slower and less safe. FCP-MPC moves the expensive calibration offline, so at runtime safety is a fieldwise query against a cached bound—fast, largely obstacle-count-insensitive, and predictable.

TABLE I

2D NAVIGATION RESULTS ON ETH-UCY PEDESTRIAN DATASETS. ALL METRICS ARE AVERAGED ACROSS SCENES; LOWER IS BETTER. THE FCP-MPC (HARD) AND FCP-MPC (SOFT) ROWS USE OUR FULL METHOD WITH THE *online-adapted* (AFCP) ENVELOPE OF (29); TABLE II ABLATES THE FIXED OFFLINE (FCP) VERSUS ONLINE-ADAPTED (AFCP) CHOICE FOR EACH CONSTRAINT MODE.

Dataset	Method	Collision rate ↓	Infeasible rate ↓	Steps to goal ↓	Ctrl. time (ms) ↓
eth	ACP-MPC [2]	0.079	0.284	79.0	0.94
	CC-MPC [3]	0.036	N/A	96.7	1.08
	ECP-MPC [5]	0.039	0.276	85.7	42.16
	FCP-MPC (hard)	0.107	0.482	83.7	2.27
	FCP-MPC (soft)	0.119	N/A	39.3	3.16
hotel	ACP-MPC [2]	0.041	0.428	80.3	1.23
	CC-MPC [3]	0.049	N/A	99.3	1.48
	ECP-MPC [5]	0.047	0.306	timeout	52.38
	FCP-MPC (hard)	0.131	0.287	73.7	3.14
	FCP-MPC (soft)	0.060	N/A	62.7	4.24
univ	ACP-MPC [2]	0.000	0.660	timeout	4.70
	CC-MPC [3]	0.053	N/A	timeout	5.78
	ECP-MPC [5]	0.204	0.530	timeout	78.27
	FCP-MPC (hard)	0.167	0.430	timeout	7.84
	FCP-MPC (soft)	0.182	N/A	99.0	12.74
zara1	ACP-MPC [2]	0.188	0.573	timeout	0.88
	CC-MPC [3]	0.041	N/A	90.0	1.07
	ECP-MPC [5]	0.098	0.180	82.0	49.16
	FCP-MPC (hard)	0.233	0.688	81.0	1.99
	FCP-MPC (soft)	0.174	N/A	62.3	3.34
zara2	ACP-MPC [2]	0.262	0.628	94.0	1.09
	CC-MPC [3]	0.056	N/A	timeout	1.25
	ECP-MPC [5]	0.220	0.412	timeout	52.12
	FCP-MPC (hard)	0.147	0.470	timeout	2.48
	FCP-MPC (soft)	0.139	N/A	71.3	3.94

TABLE II

ABLATION OF THE ONLINE ENVELOPE ADAPTATION (29) IN 2D. WITHIN EACH CONSTRAINT MODE (HARD / SOFT), THE BETTER VALUE PER METRIC IS IN BOLD. LOWER VALUES INDICATE BETTER PERFORMANCE.

Dataset	Constraint	Online adapt.	Collision rate ↓	Infeasible rate ↓	Steps to goal ↓	Ctrl. time (ms) ↓
eth	Hard	No	0.062	0.815	83.7	0.87
	Hard	Yes	0.107	0.482	83.7	2.27
	Soft	No	0.134	N/A	39.3	2.36
	Soft	Yes	0.119	N/A	39.3	3.16
hotel	Hard	No	0.120	0.447	timeout	1.65
	Hard	Yes	0.131	0.287	73.7	3.14
	Soft	No	0.080	N/A	timeout	2.86
	Soft	Yes	0.060	N/A	62.7	4.24
univ	Hard	No	0.013	0.203	timeout	5.15
	Hard	Yes	0.167	0.430	timeout	7.84
	Soft	No	0.149	N/A	94.0	9.38
	Soft	Yes	0.182	N/A	99.0	12.74
zara1	Hard	No	0.254	0.872	81.0	0.87
	Hard	Yes	0.233	0.688	81.0	1.99
	Soft	No	0.213	N/A	57.3	2.40
	Soft	Yes	0.174	N/A	62.3	3.34
zara2	Hard	No	0.177	0.563	timeout	1.33
	Hard	Yes	0.147	0.470	timeout	2.48
	Soft	No	0.170	N/A	83.3	2.70
	Soft	Yes	0.139	N/A	71.3	3.94

c) *Certified hard filter vs. practical soft deployment.*:

The same offline field can be deployed in two ways. As a *hard* filter it carries the formal coverage guarantee of Theorems 1–2; the price is conservativeness, so its infeasible rate grows with density (0.094 at $N_{\text{obs}} = 50$ to 0.269 at 280), though it still reaches the goal at both densities at the lowest per-step cost of any method. As a *soft* penalty the field carries *no* formal guarantee, but in practice removes infeasibility entirely and attains the lowest collision rate (0.032 at 280, within

$\alpha = 0.10$) while reaching the goal in every seed. Notably, FCP-MPC stays within the target misscoverage level *while actively traversing* the dense field, whereas the safe-looking baseline (ACP) attains its low collision count only by crashing early, and ECP is both slow and unsafe. We therefore position the hard filter as the certified variant and the soft penalty as the recommended practical deployment in dense dynamic scenes; we attribute the soft variant’s reliable goal-reaching to the spatial coherence of the field-level bound, which exposes

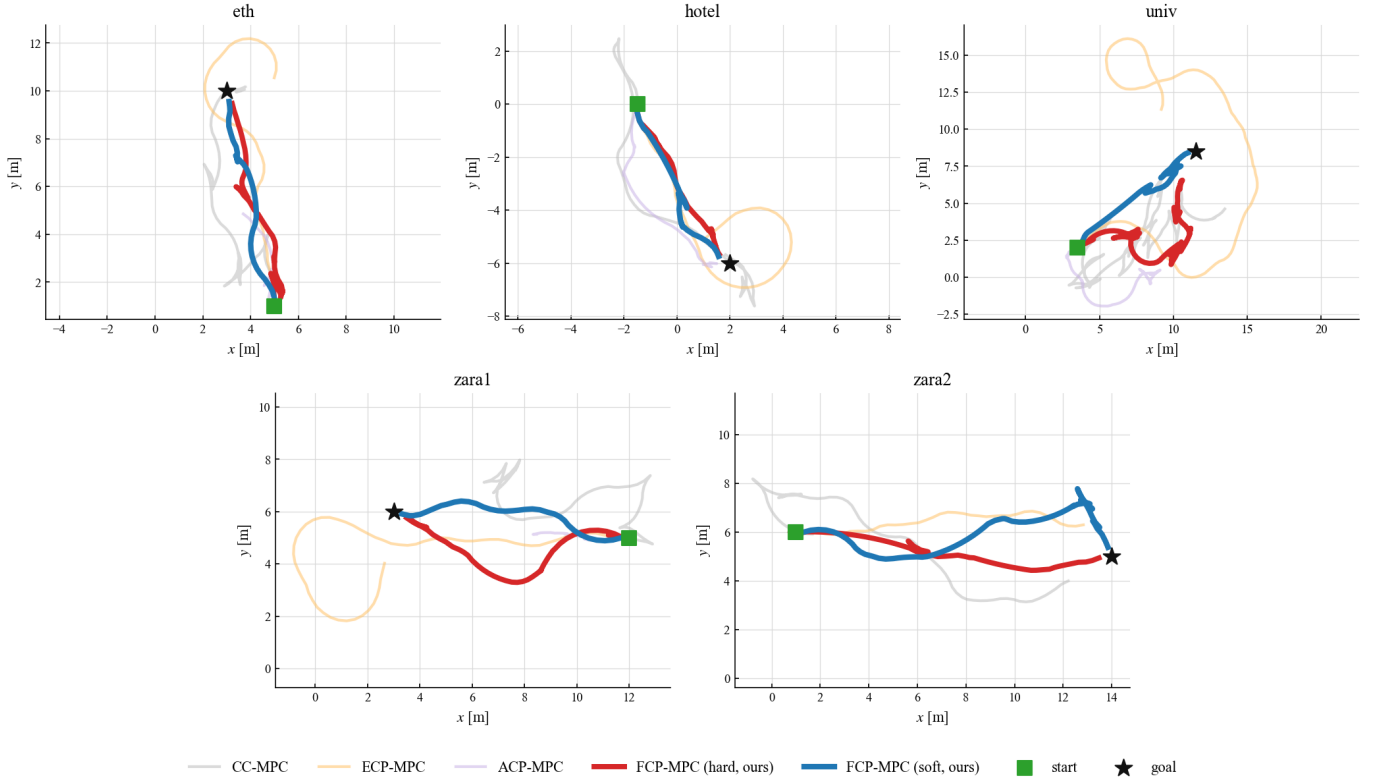


Fig. 5. Representative closed-loop trajectories on one scene of each of the five ETH-UCY datasets (green squares: start; black stars: goal). Both variants of FCP-MPC (ours)—soft (thick blue) and hard (thick red)—reach the goal along direct paths, whereas the baselines, drawn in lighter lines, tend to wander or stall: CC-MPC (gray) is slow and conservative, ECP-MPC (orange) detours, and ACP-MPC (purple) often fails to make steady progress in dense scenes such as univ.

TABLE III

3D QUADROTOR NAVIGATION RESULTS WITH 280 DYNAMIC OBSTACLES, AVERAGED OVER 17 SEEDS ($\alpha = 0.10$). FCP-MPC (HARD) AND FCP-MPC (SOFT) DEPLOY THE *same* OFFLINE-CONFORMALIZED FIELD AS A HARD FEASIBILITY FILTER AND AS A SOFT PENALTY, RESPECTIVELY. STEPS-TO-GOAL IS AVERAGED OVER SUCCESSFUL EPISODES; “CRASHED” / “TIMEOUT” MARK METHODS THAT NEVER REACH THE GOAL.

Method	Collision rate ↓	Infeasible rate ↓	Steps to goal ↓	Ctrl. time (ms) ↓
ACP-MPC [2]	0.032	0.130	crashed	216.9
CC-MPC [3]	0.064	N/A	186.5	215.3
ECP-MPC [5]	0.090	0.342	crashed	306.4
FCP-MPC (hard)	0.044	0.269	123.9	51.5
FCP-MPC (soft)	0.032	N/A	84.8	159.9

TABLE IV

3D QUADROTOR NAVIGATION AT A LOWER OBSTACLE COUNT ($N_{\text{obs}} = 50$), AVERAGED OVER 17 SEEDS ($\alpha = 0.10$); SAME SETUP AND COLUMNS AS TABLE III. REPORTING BOTH DENSITIES SHOWS HOW EACH METHOD BEHAVES AS THE WORKSPACE BECOMES LESS CLUTTERED.

Method	Collision rate ↓	Infeasible rate ↓	Steps to goal ↓	Ctrl. time (ms) ↓
ACP-MPC [2]	0.008	0.002	crashed	42.2
CC-MPC [3]	0.019	N/A	183.6	40.8
ECP-MPC [5]	0.059	0.107	crashed	198.8
FCP-MPC (hard)	0.012	0.094	87.8	25.9
FCP-MPC (soft)	0.015	N/A	74.4	38.2

globally safe corridors rather than forcing locally conservative, point-wise reasoning.

X. CONCLUSION

We presented FCP-MPC, a functional conformal prediction framework that conformalizes a high-confidence lower bound

of the predicted signed distance field and enforces it as a tightened safety constraint within a sampling-based model predictive controller. By conformalizing uncertainty at the level of the spatial *field*—rather than at individual states or trajectories—the method reasons about safety globally and coherently, and its offline-online decomposition keeps the

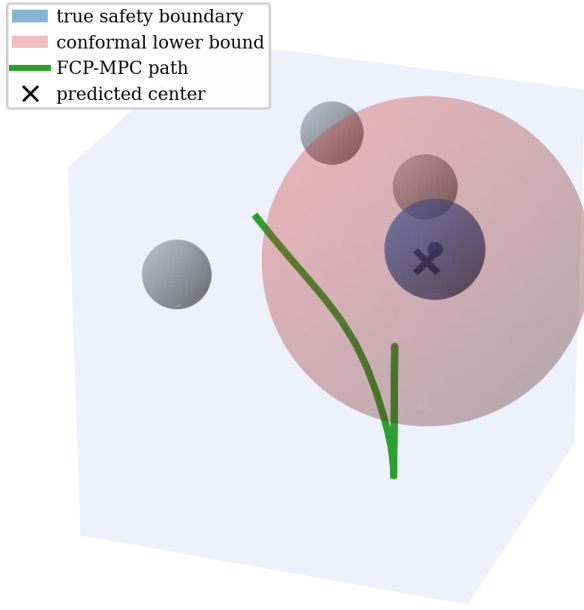


Fig. 6. True and conformal safety bounds in 3D, zoomed onto a single obstacle. The solid blue ball is the *true* safety boundary $D_{t+i}(x) = r_{\text{safe}}$ around the actual obstacle (used only for evaluation); the translucent red shell is the *conformal lower-bound* boundary $\underline{D}_{t+i|t}(x) = r_{\text{safe}}$ obtained from the functional conformal prediction, i.e. the nominal safety region around the predicted center inflated by the margin. The conformal surface encloses the true one, certifying coverage, and the executed FCP-MPC path (green) threads just outside it—progressing efficiently while respecting the worst-case safety constraint. Faint gray spheres are other obstacles in the scene.

per-step computation low: a GMM-based envelope is fitted offline, while online only a low-dimensional coefficient vector is refined by a lightweight adaptive update. We established asymptotic closed-loop safety both under exchangeability and, through the online adaptation, under distribution shift.

Across the ETH-UCY pedestrian benchmarks and a dense 3D quadrotor task with up to 280 moving obstacles, FCP-MPC attained a favorable balance of safety, feasibility, and efficiency, scaling far more gracefully than online uncertainty-reasoning baselines—with its certified hard variant the only method to stay within the real-time budget in the densest 3D scenes.

Limitations and future work. Because the conformalized envelope bounds the worst case, the hard-constrained variant remains conservative and can still report a non-zero infeasible rate in the densest 2D scenes; the soft and online-adapted variants mitigate, but do not eliminate, this trade-off. Our guarantees further assume a bounded low-level tracking error and a *fixed* functional basis. Relaxing these assumptions—learning the basis online, folding perception and state-estimation error into the same conformal pipeline, and extending the fieldwise conformal prediction to richer robot and obstacle dynamics—are promising directions, as are tighter integration with full-stack perception and validation on physical hardware.

DERIVATION OF $\check{U}_{t+i|t}$

a) *Exact union-of-ellipsoids representation.* (will be moved to appendix): Because g is a maximum over mixture components, $\mathcal{T}_{t+i|t}$ decomposes exactly as a union of component-wise ellipsoids (cf. Eq. (9) in [23]):

$$\mathcal{T}_{t+i|t} = \bigcup_{k=1}^K \mathcal{T}_{t+i|t}^{(k)}, \quad (38)$$

$$\mathcal{T}_{t+i|t}^{(k)} := \left\{ \xi : \varphi(\xi; \hat{\mu}_k, \hat{\Sigma}_k) \geq \frac{\lambda_i}{\hat{\pi}_k} \right\}.$$

Each $\mathcal{T}_{t+i|t}^{(k)}$ is an ellipsoid

$$\mathcal{T}_{t+i|t}^{(k)} = \left\{ \xi : (\xi - \hat{\mu}_k)^\top \hat{\Sigma}_k^{-1} (\xi - \hat{\mu}_k) \leq r_{k,i}^2 \right\}, \quad (39)$$

where the radius is

$$r_{k,i}^2 = \max \left\{ 0, -2 \log \left(\frac{\lambda_i}{\hat{\pi}_k} (2\pi)^{p_i/2} \sqrt{\det(\hat{\Sigma}_k)} \right) \right\}.$$

b) *From coefficient regions to a functional upper envelope:* Let the discretized basis vector at coordinate x be $\varphi_i(x) \in \mathbb{R}^{p_i}$ (in code, $\varphi_i(\cdot)$ is represented implicitly by the PCA components ϕ_i over the discretized domain). For coefficients $\xi \in \mathbb{R}^{p_i}$, the projected value is $\xi^\top \varphi_i(x)$. Using (38), we finally define the upper bound $\check{U}_{t+i|t}(x)$ as

$$\check{U}_{t+i|t}(x) := \sup_{\xi \in \mathcal{T}_{t+i|t}} \xi^\top \varphi_i(x) = \max_{1 \leq k \leq K} \sup_{\xi \in \mathcal{T}_{t+i|t}^{(k)}} \xi^\top \varphi_i(x).$$

For an ellipsoid (39), the inner supremum is given in closed form:

$$\sup_{\xi \in \mathcal{T}_{t+i|t}^{(k)}} \xi^\top \varphi_i(x) = \hat{\mu}_k^\top \varphi_i(x) + r_{k,i} \left(\varphi_i(x)^\top \hat{\Sigma}_k \varphi_i(x) \right)^{1/2}. \quad (40)$$

In implementation, we add a small diagonal jitter to $\hat{\Sigma}_k$ for numerical stability.

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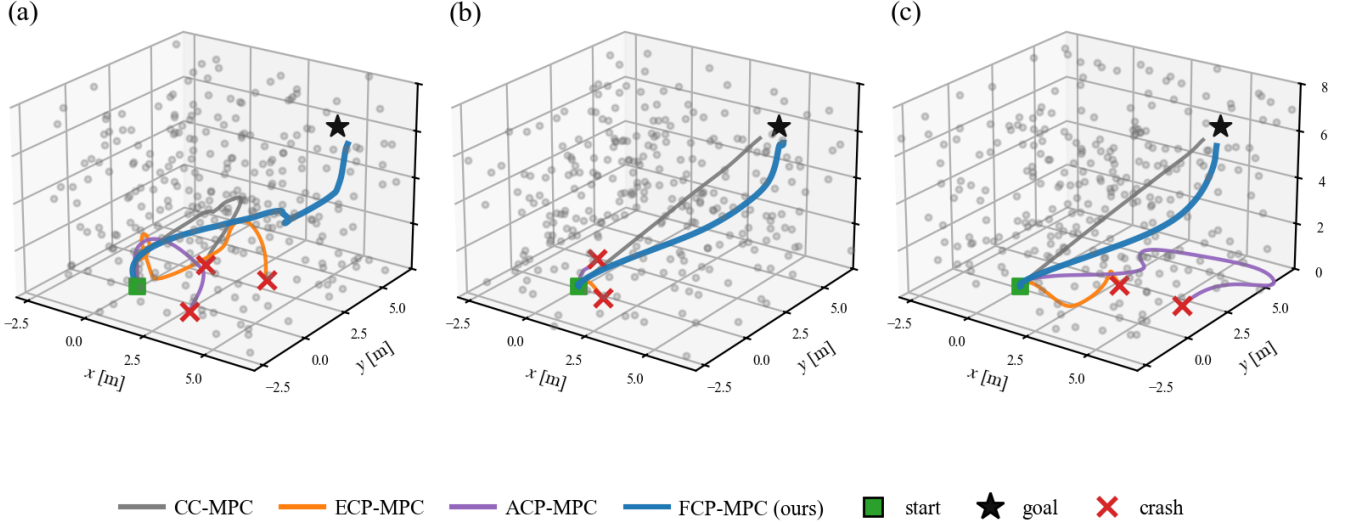


Fig. 7. Closed-loop trajectory comparison in the dense 3D quadrotor task ($M=280$ dynamic obstacles) across three random seeds, shown in panels (a)–(c). Each panel overlays all four controllers—CC-MPC (gray), ECP-MPC (orange), ACP-MPC (purple), and FCP-MPC (ours, blue). Green square: start; black star: goal; a red $\times$ marks where a controller lost control and fell to the floor. FCP-MPC is the only method that makes consistent progress to the goal across seeds, while the baselines stall or crash early in the dense field.

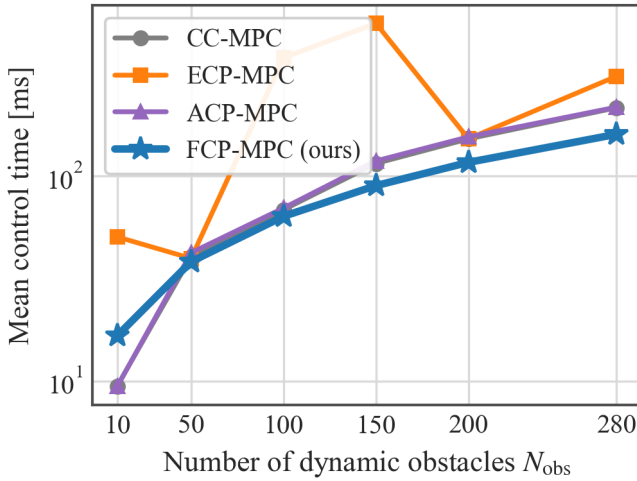


Fig. 8. Scalability comparison of MPC methods in dense 3D environments. The figure reports the mean control computation time per planning step as a function of the number of dynamic obstacles.

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